Logic and Set Theory

Flag-style derivations

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TU/e

An example of a mathematical proof

Theorem
For all $x \in \mathbb{Z}$, if x is even, then so is x^2.

Proof:
Let $x \in \mathbb{Z}$, and suppose x is even;
we need to prove that x^2 is even too.

If x is even, then $x = 2y$ for some $y \in \mathbb{Z}$.

Then $x^2 = (2y)^2 = 4y^2 = 2(2y^2)$ and $2y^2 \in \mathbb{Z}$.

So x^2 is even. □

(sub)goal

generating hypothesis

pure hypothesis

conclusion

Abstraction from concrete content



Aristotle (384-322 BC)

First study principles of logical reasoning in the abstract context of propositional and predicate logic.

Goal: a formal framework of derivation that formalises the logical principles that we usually apply in mathematical proofs.

(By comparison: when we perform a complicated mathematical calculation, we use basic facts about addition and multiplication.)

Exposing logical structure

Theorem
For all $x \in \mathbb{Z}$, if x is even, then so is x^2.

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If x is even, then $x = 2y$ for some $y \in \mathbb{Z}$.

Then $x^2 = (2y)^2 = 4y^2 = 2(2y^2)$ and $2y^2 \in \mathbb{Z}$.

So x^2 is even. □

goal
generating hypothesis
pure hypothesis
(intermediate) conclusion

Let $x \in \mathbb{Z}$

Suppose: x is even

$x = 2y$ for some $y \in \mathbb{Z}$

$x^2 = (2y)^2 = 4y^2 = 2(2y^2)$

$2y^2 \in \mathbb{Z}$

x^2 is even

For all $x \in \mathbb{Z}$, if x is even, then so is x^2

Inference

Let $P_1, P_2, \ldots, P_n$ and Q be propositions.

> Q is a correct conclusion from n premises $P_1, P_2, \ldots, P_n$
> if, and only if,
> $P_1 \wedge P_2 \wedge \cdots \wedge P_n \models^{val} Q$

If $n = 0$, then $P_1 \wedge P_2 \wedge \cdots \wedge P_n = \texttt{True}$ (by convention).

Note: if $\texttt{True} \models^{val} Q$, then $Q \equiv^{val} \texttt{True}$.

That is, Q is a tautology.

Inference (some examples)

	premises	correct conclusion	reason
0		$P \vee \neg P$	$\texttt{True} \models^{val} P \vee \neg P$
1	$\neg P \wedge \neg Q$	$\neg Q$	$\neg P \wedge \neg Q \models^{val} \neg Q$
		$\neg(P \vee Q)$	$\neg P \wedge \neg Q \models^{val} \neg(P \vee Q)$
2	$P \Rightarrow Q$ P	Q	$(P \Rightarrow Q) \wedge P \models^{val} Q$
	$\vdots$	$\vdots$	$\vdots$

Derivation

The formal system of flag-style derivations formalises the way we usually conduct logical reasonings.

It consists of two types of inference rules:

- elimination rules to draw correct logical conclusions from facts or hypotheses;
- introduction rules to simplify the logical structure of goals.

First, we use them to show that some formula is a tautology (a logical truth)

Later, we will also use them to conduct proofs of mathematical statements

Example

Show with a derivation (according to the methods of part II of the book) that the formula

$$(a \Rightarrow (b \wedge c)) \Rightarrow ((a \Rightarrow b) \wedge (a \Rightarrow c))$$

is a tautology.

[See the video 15 Implication and conjunction (example).]

Introduction of implication

How do we establish an implication in a reasoning?

$\Rightarrow$-introduction:

	{ Assume: }
(k)	P
	$\vdots$
$(\ell - 1)$	Q
	{ $\Rightarrow$-intro on (k) and $(\ell - 1)$: }
(ℓ)	$P \Rightarrow Q$

$(\uparrow)$

Introduction of conjunction

How do we establish a conjunction in a reasoning?

$\wedge$-introduction:

	$\vdots$
(k)	P
	$\vdots$
(ℓ)	Q
	{ $\wedge$-intro on (k) and (ℓ): }
(m)	$P \wedge Q$

$(\uparrow)$ $(k < m), (\ell < m)$

Elimination of implication

How can we use an implication in a reasoning?

$P \Rightarrow Q \models^{val} ??$

$(P \Rightarrow Q) \wedge P \models^{val} Q$

$\Rightarrow$-elimination:

$$
\begin{array}{ll}
 & |||| \\
(k) & P \Rightarrow Q \\
 & |||| \\
(\ell) & P \\
 & |||| \\
 & \{\Rightarrow\text{-elim on } (k) \text{ and } (\ell)\text{: }\} \\
(m) & Q \\
(\downarrow) & (k < m), (\ell < m)
\end{array}
$$

Elimination of conjunction

How can we use a conjunction in a reasoning?

$P \wedge Q \models^{val} \left\{ \begin{array}{l} P \\ Q \end{array} \right.$

$\wedge$-elimination:

$$
\begin{array}{ll}
 & |||| \\
(k) & P \wedge Q \\
 & |||| \\
 & \{\wedge\text{-elim on } (k)\text{: }\} \\
(\ell) & P \\
(\downarrow) & (k < \ell)
\end{array}
$$

$$
\begin{array}{ll}
 & |||| \\
(k) & P \wedge Q \\
 & |||| \\
 & \{\wedge\text{-elim on } (k)\text{: }\} \\
(\ell) & Q \\
(\downarrow) & (k < \ell)
\end{array}
$$

Proof strategy

Always work according to the following strategy:

1. write goal (i.e., formula for which you want to prove that it is a tautology) at bottom;
2. use introduction rules to logically simplify goal (this will typically generate hypotheses);
3. when a goal without further logical structure remains, use elimination rules to combine hypotheses and already established facts to prove simplified goal.

Later, we will refine this strategy a bit.

Introduction of negation

How do we establish a negation in a reasoning?

¬-introduction:

	{ Assume: }
(k)	P
	⋮
$(\ell - 1)$	`False`
	{ ¬-intro on (k) and $(\ell - 1)$: }
(ℓ)	$\neg P$

$(\uparrow)$

Elimination of negation

How can we use a negation in a reasoning?

$P \wedge \neg P \overset{val}{\models} \texttt{False}$

¬-elimination:

$$
\begin{array}{ll}
 & \| \| \| \| \\
(k) & \neg P \\
 & \| \| \| \| \\
(\ell) & P \\
 & \| \| \| \| \\
 & \{\, \neg\text{-elim on } (k) \text{ and } (\ell)\text{: } \} \\
(m) & \texttt{False} \\
(\downarrow) & (k < m), (\ell < m)
\end{array}
$$

Example

Show with a *derivation* (according to the methods of part II of the book) that the formula

$$(P \Rightarrow \neg Q) \Rightarrow (Q \Rightarrow \neg P)$$

is a tautology.

[See the video 18 Negation (example).]

Context and Validity

The context of a line is the collection of all assumptions that have this line in their scope.

A formula at some line in a derivation is valid on this, and all lines below it, *until* the scope of one of the assumptions in its context ends.

Validity condition: rules may only be applied to valid formulas.

Example: context

{ Assume: }
(1) $P \Rightarrow \neg Q$
  { Assume: }
(2) Q
    { Assume: }
(3) P
      { $\Rightarrow$-elim on (1) and (3): }
(4) $\neg Q$
      { $\neg$-elim on (2) and (4): }
(5) `False`
    { $\neg$-intro on (3) and (5): }
(6) $\neg P$
  { $\Rightarrow$-intro on (2) and (6): }
(7) $Q \Rightarrow \neg P$
{ $\Rightarrow$-intro on (1) and (7): }
(8) $(P \Rightarrow \neg Q) \Rightarrow (Q \Rightarrow \neg P)$

Context: all hypotheses (assumptions) that are *open* (flag pole has not yet ended).

Examples:

- Context of line (4): $P \Rightarrow \neg Q$, Q and P
- Context of line (6): $P \Rightarrow \neg Q$, Q
- Context of line (8): empty

Example: validity

{ Assume: }

(1) $P \Rightarrow \neg Q$

  { Assume: }

(2) Q

    { Assume: }

(3) P

      { $\Rightarrow$-elim on (1) and (3): }

(4) $\neg Q$

      { $\neg$-elim on (2) and (4): }

(5) `False`

    { $\neg$-intro on (3) and (5): }

(6) $\neg P$

  { $\Rightarrow$-intro on (2) and (6): }

(7) $Q \Rightarrow \neg P$

{ $\Rightarrow$-intro on (1) and (7): }

(8) $(P \Rightarrow \neg Q) \Rightarrow (Q \Rightarrow \neg P)$

Validity: a formula at line (k) is valid on all subsequent lines **until** a hypothesis in the context of line (k) is withdrawn.

Examples:

- Formula at line (4) is valid at (4), (5)
- Formula at line (6) is valid at (6)
- Formula at line (8) is valid at (8) and forever after!

The derivation thus proves that the formula

$$(P \Rightarrow \neg Q) \Rightarrow (Q \Rightarrow \neg P)$$

is a tautology (provided that every formula in it is obtained by a correct application of a rule).

Correctness

If you stick to the rules, the following holds for a completed derivation (no more holes):

> Every formula appearing in the derivation is a logical consequence of its context.

In particular, if the context of some formula in a completed derivation is empty, then the formula is a tautology.

Introduction of `False`

How can we establish `False` *in a reasoning?*

$P \wedge \neg P \models_{val}$ `False`

`False`-introduction:

	$\vdots$
(k)	$\neg P$
	$\vdots$
(ℓ)	P
	{ `False`-intro on (k) and (ℓ): }
(m)	`False`

$(\uparrow)$

`False`-introduction is essentially the same as $\neg$-elimination.

Subtle difference: `False`-introduction is *intended* for reasoning from bottom to top, whereas $\neg$-elimination is *intended* for reasoning from top to bottom.

In practice, we will only use $\neg$-elimination, and never use `False`-introduction!

Elimination of `False`

How can we use `False` *in a reasoning?*

`False` $\models_{val} P$

`False`-elimination:

	$\vert\vert\vert$
(k)	`False`
	$\vert\vert\vert$
	{ `False`-elim on (k): }
(ℓ)	P

$(\downarrow)$

Definition: flag-style derivation

A **(flag-style) derivation** is a proof constructed exclusively according to the rules listed in the *TABLES FOR PART II* (see pp. 375–380 of the book).

A flag-style derivation exclusively consists of applications of those rules; it may not use any other methods.

Introduction of disjunction (idea)

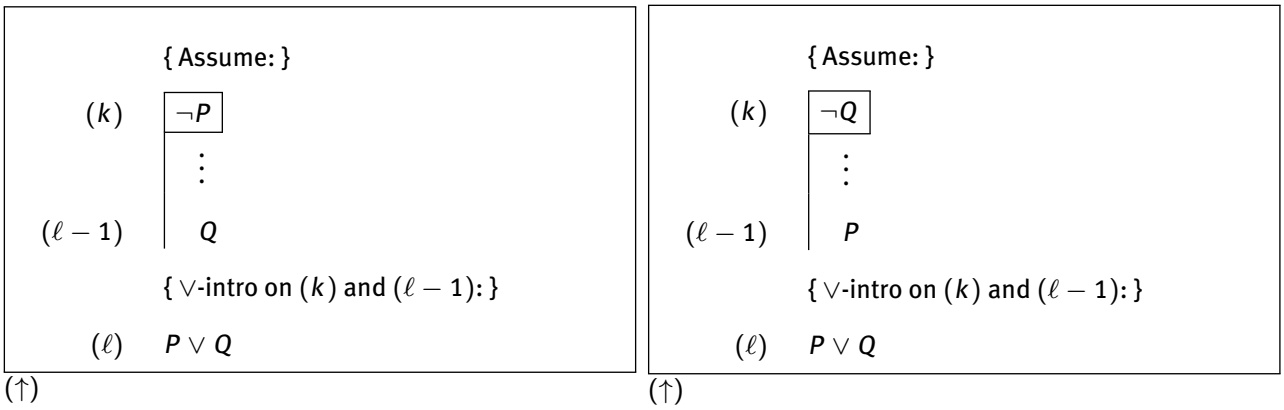
Truth table for $\vee$:

P	Q	$P \vee Q$
0	0	0
0	1	1
1	0	1
1	1	1

To establish that a disjunction $P \vee Q$ is valid, it suffices to establish that from the assumption that one of its disjuncts (P or Q) is not valid, it follows that the other disjunct is valid.

Introduction of disjunction (rule)

$\vee$-introduction:



Elimination of disjunction (idea)

Truth table for $\vee$:

P	Q	$P \vee Q$
0	0	0
0	1	1
1	0	1
1	1	1

Consider the situation that a disjunction $P \vee Q$ is valid.

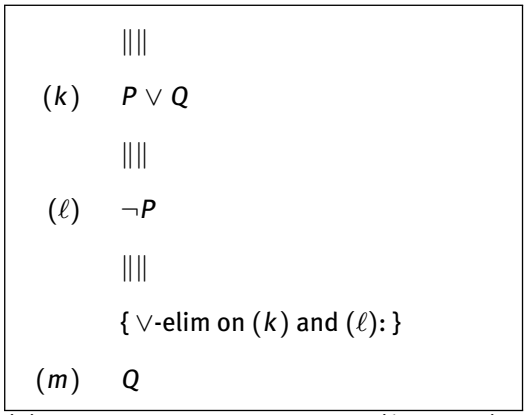
Then we cannot conclude anything about the disjuncts P and Q.

But if, in addition, one of the disjuncts, say P, is not valid, then we can conclude that the other disjunct, say Q, must be valid.

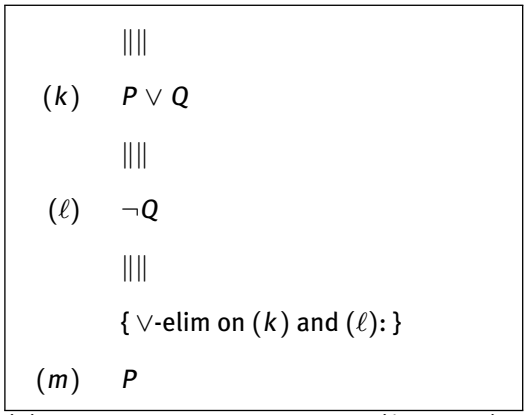
Elimination of disjunction

How do we use a disjunction in a reasoning?

∨-elimination:



(↓) $(k, \ell < m)$



(↓) $(k, \ell < m)$